

Appendix

A Supplementary Material

This appendix presents additional scheduling visualizations and changeover dynamics analyses to support the experimental findings in Section 4. We clarify the structural advantages of the proposed stability-oriented policy through machine-level details and quantitative decomposition.

A.1 Production Resource Parameters of the Jiangsu Case Plant

Table A1 summarises the three machine groups that constitute the primary production resources in the Jiangsu case plant. GT150 machines can process silicone or TPU with a daily capacity of 4,800 units, and can also operate GT130 molds via limited adapter devices. GT130 machines process TPU-only products with a daily capacity of 3,000 units. Accessories such as heat-dissipation rings or stands are produced on CNC machines (SK750) with type-dependent capacities ranging from 2,000 to 4,000 units per day. Mold changes incur a fixed downtime $h_m = 5$ hours for all molding machines.

Table A1: Production Resources in the Jiangsu Case Plant

Machine	Type	Qty	Material	Mold Type	Daily Capacity (units)
GT150	8	Silicone / TPU	GT150; GT130		4,800
GT130	4	TPU	GT130		3,000
CNC (SK750)	2	Metal accessories	Ring / Stand		2,000–4,000

The mold–machine compatibility is encoded by a binary parameter $\Gamma_{p,m}$, indicating whether product p can be processed on machine m . For notation consistency, the compatibility constraint is

$$x_{o,m,t} > 0 \Rightarrow \Gamma_{p(o),m} = 1 \quad (\text{A1})$$

where $x_{o,m,t}$ is the assigned quantity of order o on machine m at day t .

A.2 Detailed Machine-Level Gantt Charts for Scheme C

To supplement the group-level scheduling patterns in Fig. 5 of the main text, we provide machine-level Gantt charts for Scheme C across three key equipment groups: accessory machines (ADPT), Group 130 (GT130), and Group 150 (GT150), as shown in Figs. A1–A3.

All groups exhibit consistent behaviour: each machine maintains stable, unfragmented daily production blocks. This directly reflects the **one-mold-per-day** rule, which eliminates intra-day mold switching. In contrast to fragmented schedules under flexible policies, Scheme C ensures continuous operation and predictable capacity usage. The stable structure eliminates micro setup losses and supports robust schedule feasibility, explaining its 100% on-time delivery performance.

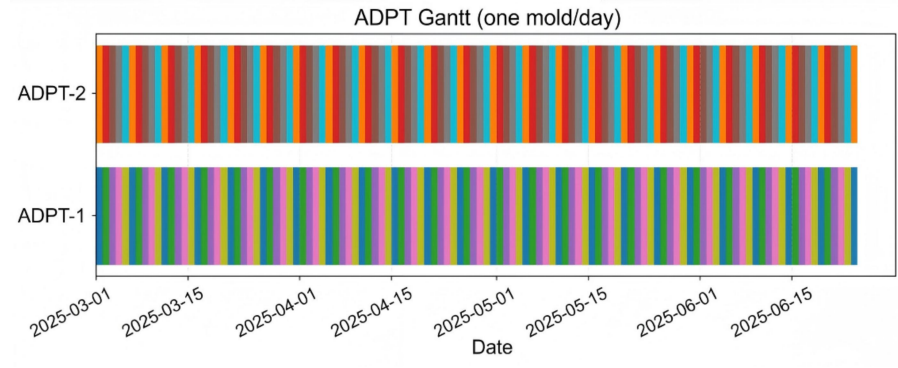


Fig. A1: Machine-level Gantt chart for accessory machines (ADPT) under Scheme C.

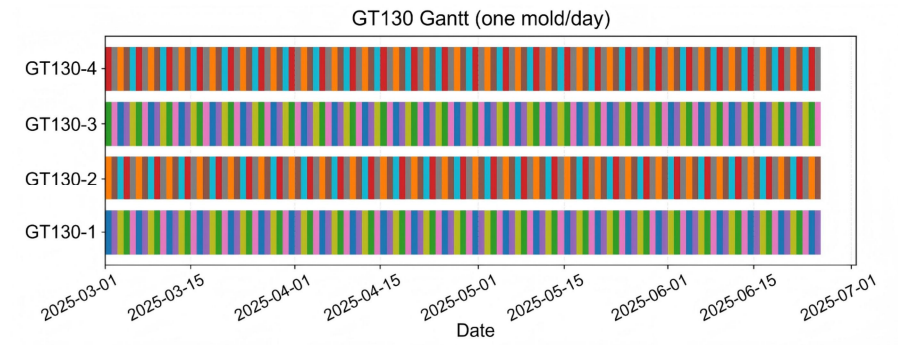


Fig. A2: Machine-level Gantt chart for Group 130 machines (GT130) under Scheme C.

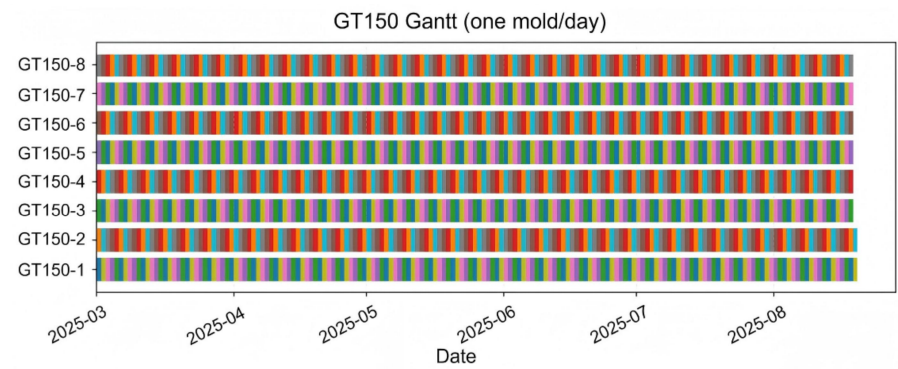


Fig. A3: Machine-level Gantt chart for Group 150 machines (GT150) under Scheme C.

A.3 Changeover Dynamics Analysis

We compare mold changeover behaviour between Scheme B and Scheme C in Fig. A4, including total switch counts and changeover-induced capacity loss.

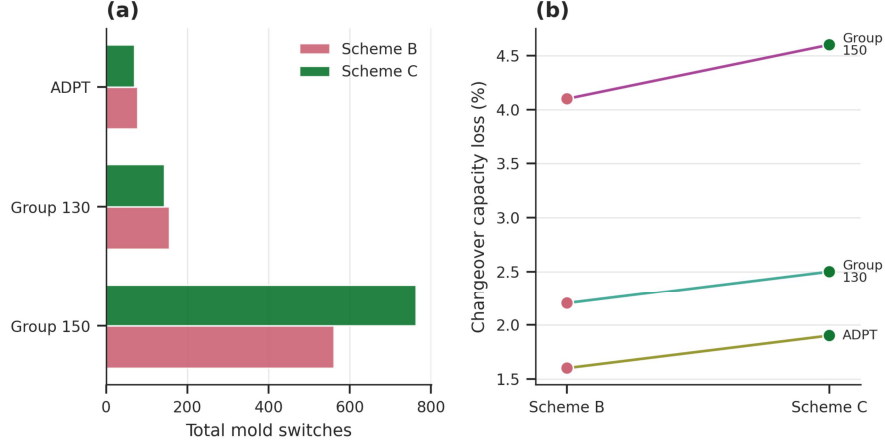


Fig. A4: Changeover dynamics comparison between Scheme B and Scheme C: (a) total mold switches; (b) capacity loss.

Scheme C yields more switches in heavily utilised groups: GT150 increases by approximately 36%, ADPT by about 25%, while GT130 remains nearly unchanged. However, capacity loss rises only marginally: roughly +0.3 percentage points in ADPT/GT130 and +0.5 percentage points in GT150, all below 5%.

The key lies in switching timing: Scheme B scatters frequent intra-day changeovers, causing fragmented and uncontrolled capacity loss. Scheme C concentrates switches at day boundaries, resulting in manageable, predictable losses despite a higher total switch count. This trade-off confirms that stability, not minimal switching, ensures delivery reliability in constrained environments.

B Detailed Mathematical Formulations for the Planning Layer

B.1 Rolling-Window Instantiation

Let the full planning horizon be $\mathcal{D} = \{1, \dots, H\}$. At each iteration, optimization runs over an active window $\mathcal{D}_{\text{act}} \subseteq \mathcal{D}$ with $|\mathcal{D}_{\text{act}}| \leq W$. Boundary states are inherited from fixed prior decisions for temporal consistency:

$$I_{f, d_{\text{start}}-1} = \hat{I}_{f, d_{\text{start}}-1}, \quad \delta_{m, k, d_{\text{start}}-1} = \hat{\delta}_{m, k, d_{\text{start}}-1}. \quad (\text{B1})$$

B.2 Linearization of Mold Changeovers

Binary variables $\Delta_{m,d}$ denote mold changes. The following constraints link $\Delta_{m,d}$ to assignment variations:

$$\Delta_{m,d} \geq \delta_{m,k,d} - \delta_{m,k,d-1}, \quad \forall m, k, d, \quad (\text{B2})$$

$$\Delta_{m,d} \geq \delta_{m,k,d-1} - \delta_{m,k,d}, \quad \forall m, k, d. \quad (\text{B3})$$

Effective capacity is defined as:

$$\tilde{c}_{m,d} = c_m - \Delta_{m,d} \cdot h_m. \quad (\text{B4})$$

B.3 Order Timing Constraints

Order production is restricted to its feasible time window:

$$q_{o,d} = 0, \quad \forall d < T_o^s \text{ or } d > T_o^e, \quad (\text{B5})$$

$$\sum_d q_{o,d} \leq Q_o, \quad \forall o \in \mathcal{O}. \quad (\text{B6})$$

These constraints enforce delivery compliance and prevent overproduction.